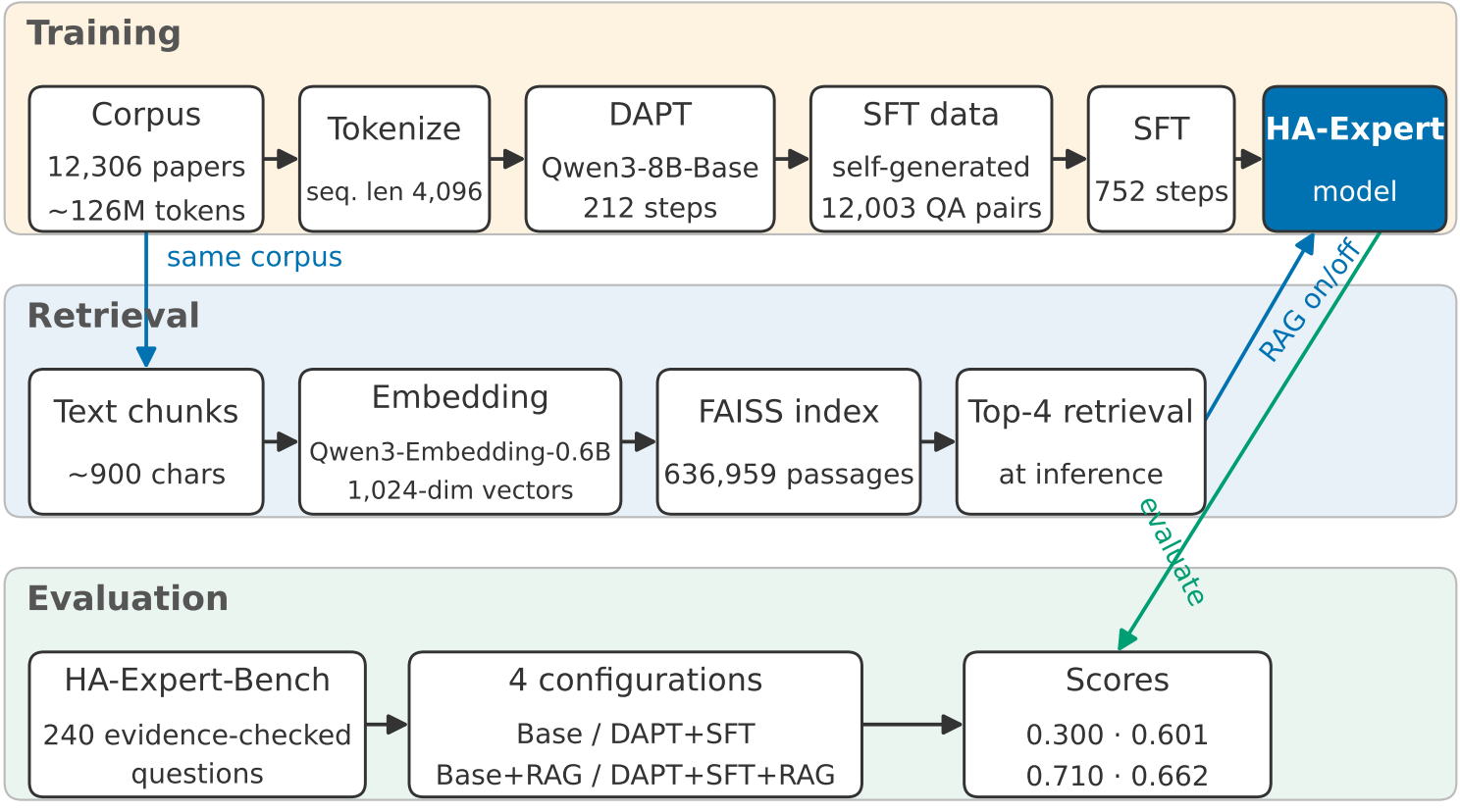
Training
Corpus
12,306 papers
~126M tokens
Tokenize
seq. len 4,096
DAPT
Qwen3-8B-Base
212 steps
SFT data
self-generated
12,003 QA pairs
SFT
752 steps
HA-Expert
model
same corpus
Retrieval
Text chunks
~900 chars
Embedding
Qwen3-Embedding-0.6B
1,024-dim vectors
FAISS index
636,959 passages
Top-4 retrieval
at inference
RAG on/off
evaluate
Evaluation
HA-Expert-Bench
240 evidence-checked questions
4 configurations
Base / DAPT+SFT
Base+RAG / DAPT+SFT+RAG
Scores
0.300 · 0.601
0.710 · 0.662